Lab 08 – Passive OT IDS with Zeek

Industrial Security (Bachelor) • Maps to Section 08 • 45–60 min

LAB 08

Goal

Show that a passive ICS-aware IDS can detect the Modbus injection from Lab 02 without ever sending a packet itself.

Step 1 – Stack and baseline traffic

The Zeek container is already up and sniffing on the lab subnet. Generate baseline traffic:

```
docker compose exec student python3 /labs/scripts/modbus_poll_loop.py &
POLL=$!
sleep 60
kill $POLL
```

Inspect host-side logs:

```
ls bachelor/labs/zeek-logs/
tail bachelor/labs/zeek-logs/modbus.log
```

Step 2 – Trigger the injection

```
docker compose exec student python3 /labs/scripts/modbus_inject.py
```

Within seconds, a notice should land in `notice.log` tagged `ModbusWatch::Unsolicited_Write`.

Step 3 – Tune one false positive

Edit `bachelor/labs/zeek/local.zeek` so writes from the student container (172.28.0.20) are not treated as notices (assume the engineering WS is sanctioned). Restart:

```
docker compose restart zeek
```

Re-run the injection – the notice should now be suppressed.

Deliverable

Hand in: `notice.log` excerpts before and after tuning, your modified `local.zeek`, and a 5-line answer to: "*Who would receive this alert in a real SOC, and what is their first action?*"